

Reading 6.7: DPO and Other Alignment Options

Welcome to this lesson on Direct Preference Optimisation (DPO)—an emerging approach that simplifies aligning AI models with human preferences. By the end of this video, you will:

- Understand the core principles of DPO
- See how it compares to Reinforcement Learning with Human Feedback (RLHF).
- Learn about additional alignment options

Due to the complexity, high resource requirements, and substantial costs associated with RLHF, researchers are actively seeking more efficient alternatives. One promising approach is Direct Preference Optimisation (DPO). Unlike RLHF, DPO aims to streamline the process of aligning AI models with human preferences by optimising directly for user preferences without the need for reinforcement learning loops.

A key challenge in RLHF arises because the output y is sampled from the language model itself. This sampling process introduces non-differentiable elements, which prevent the straightforward application of gradient descent for optimising the RLHF objective function.

Direct Preference Optimisation (DPO) is an alternative to RLHF that aligns AI models with human preferences by directly optimising the model's parameters based on pairwise comparisons of outputs. Instead of relying on a reward model and reinforcement learning, DPO finetunes the model by maximising the probability of preferred outputs over non-preferred ones.

Why it may be better than RLHF:

1. Simpler training: no reinforcement learning and reward model training
 2. Stability: avoids issues like reward hacking and unstable learning, inherent to reinforcement learning
 3. Efficiency: requires fewer computational resources and less hyperparameter tuning
- We start with looking at the RLHF objective as a constrained optimisation problem, which means that we want to maximise the reward but at the same time we want the KL divergence to be small.

$$RLHF \text{ Objective} = \max_{\pi_{\theta}} E_{prompt(x) \sim D, y \sim \pi_{\theta}(y|x)} \left[RM_{\phi}(x, y) - \beta D_{KL} \frac{\pi_{\theta}(s)}{\pi(s)} \right]$$

x: prompt/context
y: response

The RLHF objective has an analytical solution for finding the optimal policy π_* given by following expression:

$$\pi_*(y|x) = \frac{1}{Z(x)} \pi(y|x) e^{\frac{1}{\beta} r(x,y)} \quad (1)$$

$$Z(x) = \sum_y \pi(y|x) e^{\frac{1}{\beta} r(x,y)}$$

The problem however is that the term $Z(x)$ is intractable because we would have to generate all possible LLM outputs y for a given prompt x .

We are interested in finding the reward $r(x,y)$, given an optimal policy $\pi_*(y|x)$:

$$\pi_*(y|x) = \frac{1}{Z(x)} \pi(y|x) e^{\frac{1}{\beta} r(x,y)}$$

$$\log \pi_*(y|x) = \log \left[\frac{1}{Z(x)} \pi(y|x) e^{\frac{1}{\beta} r(x,y)} \right]$$

$$\log \pi_*(y|x) = \log \pi(y|x) - \log Z(x) + \log \left[e^{\frac{1}{\beta} r(x,y)} \right]$$

$$\log \pi_*(y|x) = \log \pi(y|x) - \log Z(x) + \log \left[e^{\frac{1}{\beta} r(x,y)} \right]$$

$$\log \pi_*(y|x) = \log \pi(y|x) - \log Z(x) + \frac{1}{\beta} r(x,y)$$

$$\beta \log \pi_*(y|x) = \beta \log \pi(y|x) - \beta \log Z(x) + \beta \frac{1}{\beta} r(x,y)$$

$$\beta \log \pi_*(y|x) - \beta \log \pi(y|x) + \beta \log Z(x) = r(x,y)$$

$$\beta \log \frac{\pi_*(y|x)}{\pi(y|x)} + \beta \log Z(x) = r(x,y) \quad (2)$$

Equation (2) has 2 unknowns $r(x,y)$, and $\pi_*(y|x)$ but let's assume that we have access to $\pi_*(y|x)$. Relying on the Bradley Terry model we know that:

$$P(y^w > y^l) = \sigma[r(x, y^w) - r(x, y^l)]$$

$$\begin{aligned} \text{---} \rightarrow (2) &= \sigma \left[\beta \log \frac{\pi_*(y^w|x)}{\pi(y^w|x)} + \cancel{\beta \log Z(x)} - \beta \log \frac{\pi_*(y^l|x)}{\pi(y^l|x)} - \cancel{\beta \log Z(x)} \right] \end{aligned}$$

The intractable factor $Z(x)$ cancels out and our loss function for DPO becomes:

$$Loss_{DPO}(\pi_{\theta}:\pi_{ref}) = -E_{prompt(x), (y^w, y^l) \sim D} [\log \sigma[\beta \log \frac{\pi_*(y^w|x)}{\pi(y|x)} - \beta \log \frac{\pi_*(y^l|x)}{\pi(y|x)}]]$$

The expected value term E reflects that we take the average loss over all datapoints in our dataset of preferences D . Instead of optimising a reward model, we optimise π_* that depends on the optimal policy that depends on the optimal reward function through expression (2). The loss is differentiable and we can rely on gradient descent to find the optimal values for θ .

Other Alignment Options

Constitutional AI is pioneered by Anthropic and uses a predefined set of principles—called a ‘constitution’—to guide the LLM’s behaviour and ensure its outputs are safe, ethical, and aligned with societal norms. The methodology however requires carefully crafted guidelines that may not adapt well to more complex human preferences.

Few-Shot Prompting is a fast, cheap and leverages the in-prompt/in-context learning feature of LLMs. A carefully engineered prompt provides a few examples at inference time, guiding the output. This performance depends heavily on prompt quality.

In Adversarial Training an additional model (adversary) is used to test the LLM by generating challenging examples designed to expose misaligned behaviour. Adversarial training improves model robustness but it is resource-intensive.

Human-in-the-Loop (HITL) Evaluation is continuous monitoring and feedback from human evaluators. The model is updated periodically based on new human feedback to improve alignment iteratively. HITL evaluation is dynamic and adaptable to new preferences, but costs are high.

Try it Yourself

Now that you understand how DPO works, see if you can find recent deployments or breakthroughs that suggest that DPO might replace RLHF in aligning AI models with human preferences in the near future.
